

Vishal Kumar Allam

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PROFESSIONAL SUMMARY

- Results-driven **Software Quality / AI Model Validation Engineer** with experience evaluating enterprise AI/ML systems, data-driven applications, and production software through reproducible experimentation, quantitative analysis, and structured technical validation.
- Hands-on experience validating **machine learning, Generative AI, Large Language Models (LLMs), and Retrieval-Augmented Generation (RAG)** applications across intended use, data quality, implementation, model behavior, performance, controls, and limitations.
- Strong Python and data science expertise using **Python, Pandas, NumPy, scikit-learn, SQL, statistical analysis, experimental design, model evaluation, and reproducible testing frameworks**.
- Experienced evaluating **prompt behavior, retrieval quality, groundedness, hallucination risk, response consistency, robustness, safety, and guardrail effectiveness** through automated testing, benchmarking, red teaming, and failure-mode analysis.
- Hands-on experience with **embeddings, vector search, RAG pipelines, NLP, LLM evaluation, responsible AI, model monitoring, data quality, model deployment, and production-performance analysis**.
- Experienced reviewing source code, notebooks, model documentation, test evidence, monitoring results, data pipelines, and implementation controls to identify weaknesses and provide evidence-based technical conclusions.
- Strong technical communication skills with experience creating **validation reports, technical appendices, issue statements, quality summaries, dashboards, and executive-ready presentations** for engineers, Product teams, and leadership.

EDUCATION

University of Missouri Kansas City

"Master of Science" in Computer Science

Kansas City, MO

Jan. 2023 – May 2024

Jawaharlal Nehru Technological University

"Bachelor of Technology" in Computer Science

Hyderabad, India

Aug. 2015 - May 2019

TECHNICAL SKILLS

- **Programming & Data Science:** Python, SQL, Pandas, NumPy, scikit-learn, Jupyter Notebook, Statistical Analysis, Experimental Design
- **AI/ML Validation:** Model Validation, Model Evaluation, Benchmarking, Error Analysis, Performance Analysis, Robustness Testing, Reliability Testing, Regression Detection, Effective Challenge
- **Generative AI & NLP:** Large Language Models (LLMs), Generative AI, NLP, Prompt Testing, Prompt Engineering, Embeddings, Vector Search, Retrieval-Augmented Generation (RAG)
- **LLM/RAG Quality:** Retrieval Quality, Groundedness, Hallucination Detection, Response Consistency, Guardrail Validation, Red Teaming, Safety Evaluation, Failure-Mode Analysis
- **Responsible AI & Governance:** Model Risk Management, AI Governance, Model Controls, Model Limitations, Responsible AI, Model Inventories, Production Monitoring, Periodic Model Review
- **Engineering & MLOps:** Git, Docker, Containers, AWS, Azure, REST APIs, CI/CD, Model Deployment, Data Quality, Reproducible Testing, Production Monitoring
- **Communication & Documentation:** Validation Reports, Technical Appendices, Issue Statements, Executive Presentations, Technical Documentation, Model Findings, Risk Communication

EXPERIENCE

Test Automation Engineer *PCMI LLC (Full Time)*

Oct 2024 – Present
Dallas, TX, USA

- Designed reproducible testing and validation frameworks using Python to independently evaluate AI/ML-enabled application behavior, implementation choices, data quality, performance, controls, robustness, and production reliability.
- Evaluated **Generative AI and LLM-powered workflows** by testing prompt behavior, response consistency, groundedness, hallucination risk, safety constraints, and guardrail effectiveness across normal, adversarial, and edge-case scenarios.
- Built evaluation workflows for **RAG applications**, analyzing embeddings, vector-search results, retrieval relevance, context quality, grounding, and downstream response accuracy to identify weaknesses in retrieval and generation components.
- Applied statistical and machine learning methods using Python, Pandas, NumPy, and scikit-learn to benchmark system behavior, compare alternatives, analyze error distributions, and interpret model-evaluation results.
- Reviewed source code, APIs, notebooks, data-processing logic, technical documentation, test evidence, and production-monitoring outputs to identify implementation gaps, model limitations, and quality risks.
- Performed structured failure-mode analysis and red-team testing, investigating unexpected outputs, inconsistent behavior, degraded performance, malformed inputs, and control failures while documenting reproducible evidence.
- Evaluated ongoing production monitoring, quality metrics, drift indicators, regression results, and operational controls to identify emerging issues and incorporate findings into reusable offline validation frameworks.
- Prepared clear validation reports, technical findings, issue statements, dashboards, and executive-ready summaries while collaborating with developers, data scientists, Product teams, and technical stakeholders to constructively challenge assumptions.

Environment: Python, Pandas, NumPy, scikit-learn, Generative AI, LLMs, RAG, NLP, Embeddings, Vector Search, Prompt Testing, Groundedness, Hallucination Detection, Guardrails, Red Teaming, Statistical Analysis, Model Monitoring, REST APIs, Git, Docker, AWS, CI/CD, Responsible AI, Model Validation

SDET *Client: CVS*

Dec 2023 – Sept 2024
Remote, USA

- Validated AI/ML-enabled healthcare applications by independently assessing system requirements, data inputs, implementation logic, model behavior, controls, performance, and limitations across production-oriented workflows.
- Developed Python-based evaluation frameworks for **LLM and Generative AI applications**, measuring response quality, prompt sensitivity, hallucination risk, consistency, robustness, safety behavior, and compliance with expected business rules.
- Evaluated healthcare-focused **RAG pipelines** by testing document retrieval, embeddings, vector-search relevance, context completeness, groundedness, and answer quality across representative and adversarial scenarios.
- Applied statistical analysis and model-evaluation techniques to compare model versions, benchmark results, identify error patterns, quantify performance differences, and investigate anomalous behavior.
- Reviewed application code, API implementations, notebooks, healthcare datasets, model documentation, test results, and monitoring evidence to evaluate technical design and identify potential quality or control weaknesses.
- Performed red-team, negative, and edge-case testing to identify unsafe responses, unsupported conclusions, retrieval failures, inconsistent outputs, and guardrail weaknesses before production releases.
- Monitored production quality signals and regression trends, investigating emerging performance issues and feeding production findings back into automated validation, regression, and model-testing coverage.
- Collaborated with engineers, data scientists, Product Owners, and healthcare stakeholders to document model findings, technical limitations, risks, validation results, and recommendations in clear technical and leadership-facing formats.

Environment: Python, Pandas, NumPy, scikit-learn, Machine Learning, Generative AI, LLMs, RAG, Embeddings, Vector Search, Prompt Evaluation, Model Benchmarking, Statistical Analysis, Healthcare Data, REST APIs, SQL, Git, Docker, AWS, Model Monitoring, Responsible AI, Data Quality

Software Engineer *Infor*

Sept 2021 – Dec 2022
Hyderabad, India

- Developed Python and Java-based validation utilities for enterprise applications and data-driven systems, creating reproducible test frameworks for functional, integration, API, and backend evaluation.
- Used Python, SQL, and statistical techniques to analyze application data, validate processing logic, identify anomalies, compare expected versus observed behavior, and investigate complex software failures.
- Designed structured experiments and regression scenarios to evaluate system reliability, implementation changes, error conditions, boundary behavior, and performance across multiple application releases.
- Reviewed source code, API specifications, technical documentation, database logic, and implementation details to understand unfamiliar systems and identify areas requiring deeper validation.
- Performed data-quality validation across relational databases and APIs, testing data completeness, consistency, transformations, business rules, and downstream application behavior.
- Developed automated analysis and reporting workflows to summarize execution results, defect trends, performance observations, and quality risks across enterprise software releases.
- Integrated reproducible automated validation into Git and Jenkins-based CI/CD pipelines, supporting consistent regression testing and evidence-based release decisions.
- Collaborated with engineers, Product Managers, and technical stakeholders to investigate failures, challenge implementation assumptions, document findings, and communicate quality recommendations clearly.

Environment: Python, Java, SQL, Pandas, Data Analysis, Statistical Analysis, REST APIs, Data Quality, Reproducible Testing, Regression Analysis, Git, Jenkins, CI/CD, Technical Documentation, Root Cause Analysis, Agile Scrum

Associate Engineer QA

June 2019 – August 2021

Virtusa

Hyderabad, India

- Developed reusable automation and validation frameworks for enterprise web, mobile, API, and backend applications using Java, Python, Selenium WebDriver, Cucumber BDD, Appium, and TestNG.
- Designed systematic test scenarios covering functional behavior, regression risk, boundary conditions, negative cases, integration behavior, and system reliability across enterprise software releases.
- Performed structured error analysis using application logs, API responses, SQL data, test results, and execution traces to reproduce failures and identify underlying technical causes.
- Validated data quality and application behavior using SQL and API testing, checking transaction accuracy, business-rule consistency, backend processing, and cross-system data integrity.
- Reviewed requirements, technical documentation, service interfaces, and implementation behavior to translate open-ended quality questions into repeatable test scenarios and defensible findings.
- Created automated regression suites and repeatable test environments through Git, Jenkins, Maven, and CI/CD processes to improve software-quality monitoring and release confidence.
- Prepared detailed test reports, defect analyses, quality summaries, risk assessments, and release recommendations for engineering and project stakeholders.
- Collaborated with Software Engineers, Business Analysts, QA leads, and Product teams to challenge assumptions, investigate defects, improve testability, and strengthen overall software controls.

Environment: Java, Python, Selenium WebDriver, Appium, Cucumber BDD, TestNG, REST APIs, SQL, Data Validation, Regression Testing, Error Analysis, Git, Jenkins, Maven, CI/CD, Technical Reporting, Root Cause Analysis, Agile Scrum

AWARDS & ACHIEVEMENTS

- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.
- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded “Software Process Achievement” Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.